

KATHIRAVAN V

B.E. Electronics and Communication Engineering | 2nd Year Undergraduate | Research-Oriented Engineering Student

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ACADEMIC PROFILE

Second-year Electronics and Communication Engineering undergraduate at Panimalar Engineering College with hands-on experience developing embedded, IoT, healthcare-technology, and data-driven engineering projects. Interested in research at the intersection of embedded systems, intelligent sensing, IoT, signal processing, and applied AI. Experienced in translating engineering problems into working prototypes using microcontrollers, sensors, software, and cloud-connected systems. Seeking research exposure in an interdisciplinary environment where experimental work, technical learning, and problem solving can be developed under academic mentorship.

ACADEMIC BACKGROUND

Bachelor of Engineering (B.E.) Electronics and Communication Engineering	Panimalar Engineering College, Chennai, India	2025 – 2029 Currently in 2nd Year
Higher Secondary Certificate (HSC) Science / Higher Secondary	Saint Vincent Pallotti Matriculation Higher Secondary School, Chennai, India	Completed

ACADEMIC, RESEARCH AND PROJECT EXPERIENCE

RYDEX Smart Helmet

Embedded Systems | ESP32 | Sensors | GNU Octave

- Developed a safety-oriented embedded prototype integrating helmet-wear detection, alcohol sensing, alert mechanisms, and ignition-control logic.
- Worked with ESP32-based sensing and actuator interfaces and explored signal acquisition and visualization using GNU Octave.
- Focused on real-time sensor data, threshold-based decision logic, hardware interfacing, and system-level debugging.

AXON / HealthSync – Patient Memory Layer

Healthcare Technology | AI | Supabase | Web Application

- Designed a healthcare-oriented software architecture for organizing longitudinal patient information and supporting patient–doctor workflows.
- Explored structured patient-data handling, AI-assisted orchestration, and practical interfaces for healthcare information systems.
- Built the project with a research-oriented focus on continuity of information and technology-assisted clinical workflows.

Replenova – AI-Assisted Supply Chain Risk Dashboard

Data Systems | AI | React | TypeScript

- Developed a supply-chain risk visualization concept connecting cyclone-related port disruption to supplier delay, inventory risk, stockout, and replenishment decisions.
- Implemented an interactive dashboard to represent cascading operational effects and support data-driven decision making.
- Explored how predictive or AI-assisted recommendations can be integrated into engineering management workflows.

Panimalar Smart Bus Transport System

IoT / Web Systems | ESP32/Hardware | Supabase | Leaflet

- Developed a campus transport platform connecting student, driver, and administrator workflows.
- Integrated real-time location-oriented functionality, database-backed records, and responsive interfaces for operational use.
- Worked across hardware/software boundaries and considered reliability, usability, and deployment constraints.

RESEARCH INTERESTS

Embedded Systems • Intelligent Sensing • IoT • Signal Processing • Edge AI • Applied Machine Learning • Healthcare Technology • Cyber-Physical Systems • Sensor Data Analytics • Smart Systems

AREAS OF EXPERTISE

Embedded systems, intelligent sensing, IoT, signal processing, edge AI, applied machine learning, healthcare technology, cyber-physical systems

TECHNICAL SKILLS

Embedded & Hardware	ESP32, Arduino, sensor interfacing, GPIO, actuators, basic embedded debugging
Programming	Python, C++, Java (basic), TypeScript
Data / AI	Data processing, dashboard development, AI-assisted application concepts, signal visualization
Software	React, Vite, Tailwind CSS, Supabase, Git, GitHub, Visual Studio Code
Engineering	Problem solving, system design, debugging, prototyping, hardware–software integration

ACADEMIC AND TECHNICAL ACTIVITIES

- Participated in technical workshops, coding challenges, engineering events, and hands-on training.

- Developed multiple independent and academic prototypes spanning embedded systems, IoT, healthcare technology, and software engineering.
- Worked in team-based project environments involving system planning, implementation, testing, debugging, and presentation.
- Continuously building programming and engineering skills through project-based experimentation.

CERTIFICATIONS AND TRAINING

AI Tools Workshop Certification • GUVI Certification (HCL Collaboration) • Prompt Engineering Fundamentals • Introduction to IoT • MATLAB Basics

LANGUAGES

English – Fluent **Tamil** – Native

Note: No publications or formal research appointments are listed because none were provided in the source material.